

Finding a Three-Way Collision with Bounded Memory

Three Branching-Program Conjectures

Status. Statement: AI-written, not yet formalized.

Category. Symmetric-Key Cryptography.

Setting. Throughout, $N \geq 2$ and $f: [N] \rightarrow [N]$ is a uniformly random function, available only through an oracle answering $x \mapsto f(x)$. A k -collision is a set of k distinct points of $[N]$ with a common image under f ; a 3-collision is the case $k = 3$, and two 3-collisions are *disjoint* if they share no point. Asymptotic notation $\tilde{O}, \tilde{\Omega}, \tilde{\Theta}$ suppresses factors polylogarithmic in N . Success probabilities are over the choice of f and any coins of the algorithm; randomisation need not be allowed for separately, since the input is already random.

What is known. Four facts fix the landscape.

- (F1) *The query threshold is $N^{2/3}$.* Among q queried points the expected number of 3-collisions is $\binom{q}{3}N^{-2}$, so $q = \Theta(N^{2/3})$ queries are necessary and sufficient for constant success probability (Suzuki, Tonien, Kurosawa and Toyota, ICISC 2006). Quantitatively, an algorithm making T queries and then naming three points succeeds with probability $O((T^3 + 1)N^{-2})$. This holds for *any* algorithm, at any memory bound, and no conjecture below contradicts it.
- (F2) *Without a table, $\Theta(N)$ queries suffice.* Find a 2-collision $\{u, v\}$ by cycle-finding, retain u, v and $y = f(u)$, then sample fresh points until one maps to y . This uses $O(\log N)$ bits of memory. Between **F1** and **F2** lies a gap of $N^{1/3}$, and the conjectures below assert that memory is what closes it.
- (F3) *The interpolating algorithms.* Joux and Lucks (ASIACRYPT 2009) achieve $T = \tilde{O}(N/S)$ using a table of S collision pairs, for every $S \leq N^{1/3}$; Guo, Nan and Yao (ASIACRYPT 2025) find C pairwise disjoint 3-collisions in $\tilde{O}(CN/S)$ queries, for every $S \leq \tilde{O}(C^{2/3}N^{1/3})$. Every conjecture below is tight against one of these if true.

(F4) *At $k = 2$ memory is free.* Pollard's rho method with Brent's cycle detection finds a 2-collision in $\Theta(\sqrt{N})$ queries using $O(\log N)$ bits, and that count is optimal even with unbounded memory. Consequently *every* conjecture below is false at $k = 2$, and no proof of any of them can be indifferent to the collision order.

Models. Both are the branching program of Borodin and Cook (SICOMP 1982), specialised to a random-function oracle. The state may encode anything, so one cannot say what a program *knows*.

Sequential. A *branching program* of length T and width W is a DAG on levels $0, \dots, T$ with at most W nodes per level, in which every node below level T carries a label $x \in [N]$ and has one outgoing edge for each $y \in [N]$, and every node at level T carries a label in $[N]^3$. It is run by starting at the source, querying the label of the current node, following the edge labelled by the answer, and reading the label of the sink reached; it *succeeds* if that label is a 3-collision. Its *space* is $S = \log_2 W$, counted in bits. It is *oblivious* if all nodes at a level carry the same label, so that the query schedule $w_1, \dots, w_T$ is fixed in advance, and *repetition-free* if no point labels two levels. A program asked for C pairwise disjoint 3-collisions instead of one carries labels in $[N]^{3C}$ at level T ; nothing else changes.

Parallel. A *parallel branching program* of query width q and r rounds is the same object with levels $0, \dots, r$, each node below level r carrying a label in $[N]^q$ and having one outgoing edge for each answer tuple in $[N]^q$. The total query count is $T = rq$. Oblivious and repetition-free mean as before, now of q -tuples.

The proved base. Exactly one lower bound is known in each model, and both are for oblivious programs, where the multi-pass streaming switching lemma of Dinur (EUROCRYPT 2020) applies to the answer stream. For *adaptive* programs nothing memory-sensitive is known at $k = 3$: the only available lower bound is the query bound F1, which mentions no memory at all.

Theorem BSo (oblivious sequential programs). *Every repetition-free oblivious branching program succeeding with probability at least $\frac{1}{2}$ satisfies $T \cdot (S + \lceil \log_2 N \rceil) = \tilde{\Omega}(N)$; in particular $T = \tilde{\Omega}(N)$ at width $N^{O(1)}$.*

Theorem BPo (oblivious parallel programs). *Every repetition-free oblivious parallel branching program of space S and query width q succeeding with probability at least $\frac{1}{2}$ satisfies $T \cdot (S + q \lceil \log_2 N \rceil) = \tilde{\Omega}(N)$; in particular $T = \tilde{\Omega}(N/q)$ at width $N^{O(1)}$.*

The conjectures. Each sharpens one of these, and each is tight against an algorithm.

Conjecture BSo⁺ (the oblivious sequential curve at $k = 3$). *Every repetition-free oblivious branching program succeeding with probability at least $\frac{1}{2}$ satisfies*

$$T = \tilde{\Omega}(\max\{N^{2/3}, N/\sqrt{S}\}),$$

$$\text{equivalently} \quad T \cdot S^{1/2} = \tilde{\Omega}(N) \text{ for } S \leq N^{2/3}.$$

The first clause is **F1** and carries no content; all of it is in $T \cdot S^{1/2}$. The bound is tight: an oblivious program can accumulate $P \leq S$ collision pairs at cost $\tilde{\Theta}(PN/S)$ and then hunt a third preimage of one of the P recorded values in $\tilde{\Theta}(N/P)$ further queries, both phases non-adaptively; balancing at $P = \sqrt{S}$ gives $T = \tilde{\Theta}(N/\sqrt{S})$. So a factor $\sqrt{S}$ separates **BSo** from the conjecture inside its own class. The two clauses cross at $S = N^{2/3}$, where the curve meets the query threshold and the problem closes.

Conjecture BP3' (the oblivious parallel curve, every query width). *For every q , every repetition-free oblivious parallel branching program of width $N^{O(1)}$ succeeding with probability at least $\frac{1}{2}$ satisfies*

$$T = \tilde{\Omega}(\min\{N, N^2/q^2\}).$$

This is tight at polynomial width, where two strategies are available with $O(\log N)$ bits between rounds: one holds a 2-collision and hunts q fresh points per round, costing $T = \Theta(N)$ at any q as in **F2**; the other detects a 3-collision inside a single batch, which routing permits once the width exceeds $\binom{q}{3}$, with probability $\approx q^3/N^2$ per round and so $T = \tilde{\Theta}(N^2/q^2)$. The two cross at $q = \sqrt{N}$. Nothing is conjectured beyond polynomial width, where a larger width both permits a table and permits naming more triples per batch; already at $q = \sqrt{N}$ the proved **BPo** gives $T = \tilde{\Omega}(\sqrt{N})$ against a conjectured truth of $\tilde{\Theta}(N)$, so a factor q is missing even in the oblivious class.

Conjecture BS₄-C (*C disjoint triples, adaptive programs*). *For every $C \geq 1$, every branching program that outputs C pairwise disjoint 3-collisions with probability at least $\frac{1}{2}$ satisfies*

$$T = \tilde{\Omega}(\max\{C^{1/3}N^{2/3}, CN/S\}).$$

No obliviousness is assumed, and no bound on the width. The first clause is the query threshold for C disjoint triples, by the argument of F1 applied C times; the second is the algorithm of Guo, Nan and Yao in F3, and the two clauses cross exactly where the two regimes meet, so the conjecture is tight throughout. At $C = 1$ it is the full sequential statement $T \cdot S = \tilde{\Omega}(N)$.